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Label-efficient computational tumour infiltrating lymphocyte assessment in breast cancer (ECTIL): multicentre validation in 2340 patients with breast cancer



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Summary

Background The density of stromal tumour-infiltrating lymphocytes (TILs) is a prognostic factor for patients with triple-negative breast cancer and reflects their immune response. Computational TIL assessment has the potential to assist pathologists in this labour-intensive task, because it can be quick and reproducible. However, computational TIL assessment models heavily rely on detailed annotations and use complex deep learning pipelines that pose challenges for model iterations and clinical deployment. Here, we propose and validate a fundamentally simpler deep learning-based model that is trained in only 10 min on 100 times fewer pathologist annotations.

Methods We collected whole slide images (WSIs) with TIL scores and clinical data of 2340 patients with breast cancer, including 790 patients with triple-negative breast cancer, from three cohorts in three countries (one each in the USA, UK, and Netherlands) and three randomised clinical trials in the Netherlands. Morphological features were extracted from WSIs using a pathology foundation model. Our model, label-efficient computational stromal TIL assessment (ECTIL), directly regresses the WSI TIL score from these features. We trained ECTIL on a single cohort from The Cancer Genome Atlas (n=356, ECTIL-TCGA), on only triple-negative breast cancer samples from four cohorts (n=400, ECTIL-TNBC), and on all molecular subtypes of five cohorts (n=1964, ECTIL-combined). We computed the concordance between ECTIL and the pathologist using the Pearson's correlation coefficient (r) and computed the area under the receiver operating characteristic curve (AUROC) using the pathologist TIL scores split into the clinically relevant TILs-high ($\geq 30\%$) and TILs-low ($< 30\%$) groups. We also performed multivariate Cox regression analyses on the PARADIGM cohort with complete clinicopathological variables (n=384) to assess hazard ratios for overall survival, independent of clinicopathological factors.

Findings ECTIL-TCGA showed concordance with the pathologist over five heterogeneous external cohorts ($r=0.54$ – 0.74 , AUROC 0.80 – 0.94). ECTIL-TNBC showed a higher performance than ECTIL-TCGA on the PARADIGM cohort (r 0.64 , AUROC 0.83 vs r 0.58 , AUROC 0.80), and ECTIL-combined attained the highest concordance on an external test set (r 0.69 , AUROC 0.85). Multivariate cox regression analyses indicated that every 10% increase of ECTIL-combined TIL scores was associated with improved overall survival (hazard ratio 0.85 , 95% CI 0.77 – 0.93 ; $p=0.0007$), which was independent of clinicopathological variables and similar to the pathologist score (0.86 , 0.81 – 0.92 ; $p<0.0001$).

Interpretation In conclusion, our study showed that ECTIL could score TILs on haematoxylin and eosin-stained, formalin-fixed, paraffin-embedded WSIs in a single step, attaining high concordance with an expert pathologist. Without using deep learning-based segmentation and detection pipelines, ECTIL attained similar hazard ratios to the pathologist's score in an overall survival analysis independent of clinicopathological variables. In the future, such a computational TIL assessment model could be used to pre-screen patients for prospective de-escalation trials in patients with triple-negative breast cancer or as a tool to assist pathologists and clinicians in the diagnostic investigation of patients with breast cancer. Furthermore, our model is available online under an open-source licence, allowing translational researchers to validate and use ECTIL in future studies in breast or other cancers.

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Introduction

Breast cancer is the second-leading cause of cancer-related mortality in women worldwide.¹ Research on tumour-immune interactions has identified stromal tumour-infiltrating lymphocytes (TILs) as a key prognostic and predictive biomarker, particularly in HER2-positive and triple-negative breast cancer subtypes.² In triple-negative breast cancer, TIL levels correlate with pathological complete response, disease-free-survival, and overall survival.²⁻⁴ Moreover, high TIL levels (a TIL score higher than 50% in patients older than 40 years and 75% or more in patients younger than 40 years) at baseline could indicate an excellent overall survival in stage I triple-negative breast cancer without adjuvant or neoadjuvant chemotherapy.^{5,6}

Pathologists assess TILs through visual evaluation of haematoxylin and eosin (H&E)-stained, formalin-fixed paraffin-embedded (FFPE) whole slide images (WSIs). The TIL score is defined as the area covered by mononuclear inflammatory cells in the stromal area within the invasive tumour border, following guidelines of the

International Immuno-Oncology Biomarker Working Group on Breast Cancer (TILs Working Group).⁷ Although the improved scoring process through additional training has enhanced reproducibility among pathologists, inter-observer and intraobserver variability persists as a challenge, partly due to intratumour heterogeneity. Kos and colleagues⁸ showed intraclass correlation coefficients ranging from 0.77 to 0.94 over three study groups.

Advancements in artificial intelligence (AI) and deep learning in the past decade, coupled with digitalisation of histopathological tissue into WSIs, have driven the development of computational TIL assessment methods. Furthermore, the TILs Working Group has introduced computational guidelines to address the full complexity of the scoring process, aiding the development of computational TIL assessment methods that could be adopted for clinical use.⁹ Moreover, AI-assisted assessment of TILs could potentially offer some solutions to the challenges faced by pathologists, including diminishing the burden on pathologists and pathology departments.

Research in context

Evidence before this study

We searched Google Scholar and PubMed on Nov 1, 2023 for the search terms ("Deep Learning" OR "Artificial Intelligence") AND "Pathology" AND ("lymphocytes" OR "stromal tumour infiltrating lymphocytes" OR "TILs") AND "breast cancer", and reviewed the first 30 results. No language or date restrictions were applied. We included papers that validated deep learning applications to quantify stromal tumour infiltrating lymphocytes (TILs) in invasive breast cancer samples from haematoxylin and eosin-stained whole slide images (n=11), and reviews thereof (n=1). The included studies either compared automated TIL quantification against pathologist assessments or introduced novel lymphocyte-based biomarkers aimed at improving prognostic or predictive performance. The TIL scores provided by the published methods correlate well with pathologists (Spearman's rank correlation coefficients ranging from 0.57 to 0.79 over studies and cohorts) and were shown to have prognostic value independent of other clinicopathological variables. At the same time, their prognostic performance remains lower than that of a pathologist. All externally validated methods are pipelines of multiple steps that perform segmentation and detection to combine pixel-level information into the final biomarker, trained on very large numbers of costly tissue-level and cell-level annotations by expert pathologists.

Added value of this study

To our knowledge, this is the first study that externally validates a computational TIL assessment model in early triple-negative breast cancer with a fundamentally different design without detection and segmentation deep learning pipelines. Our model is trained solely on whole slide image-level TIL scores, in contrast to models trained using pixel-level semantic segmentations and cell annotations. Our model requires only a few hundred pathologist

annotations to be trained—a reduction by two orders of magnitude compared with published methods using thousands of annotations. Despite its few training labels, simplicity, and absence of any deep learning-based segmentation mask or cell detections, the model reached similar concordance (as measured by Spearman's rank correlation coefficients ranging from 0.47 to 0.72 across other cohorts) and prognostic value independent of other clinicopathological variables in a cohort of patients younger than 40 years with node-negative, triple-negative breast cancer who had not received neoadjuvant or adjuvant chemotherapy. To our knowledge, this is the largest retrospective validation study (2340 patients with 2354 slides over six cohorts) of any computational TIL assessment method. Our model performs well across various clinical variables, molecular subtypes, and medical centres, in both biopsies and resections.

Implications of all the available evidence

Our paper shows that our simple, data-efficient model can achieve high concordance with pathologists and retain prognostic value independent of other known prognostic clinicopathological variables. Both concordance and prognostic performance are similar to existing methods that use multi-step segmentation-detection pipelines, but our model requires substantially fewer annotations. Our findings suggest that data-intensive detection and segmentation approaches might not be necessary for computational TIL assessment. Moreover, our model resolves some, but not all, limitations and challenges of existing methods. Compared with other complex, multi-step, detection segmentation pipelines, solving these known limitations and challenges requires few additional labels and simple architectural changes. Our model will be prospectively validated in a de-escalation phase 3 clinical trial in patients with low-stage, triple-negative breast cancer.

Current models typically offer explicit pixel-level segmentations and cell detection and classifications.^{10–13} Although these models show high concordance with the pathologists' TILs score and are associated with improved survival, they still face numerous pitfalls and challenges as described by Thagaard and colleagues.¹⁴ Existing computational TIL assessment models generally encounter difficulties due to various technical slide issues, and adhering to the pathologist scoring guidelines remains challenging.⁹ Moreover, these models demand large amounts of labelled data for training, which is costly and time consuming. As these models become more complex to provide pixel-level information, continuous validation of all submodules in clinical practice could become increasingly difficult. Despite providing pixel-level segmentation and cell locations while explicitly implementing scoring guidelines, these models still encounter discordant cases due to, for example, inclusion of incorrect tissue areas or cells.

In this study, we present and validate a computational TIL assessment model—label-efficient computational stromal TIL assessment (ECTIL)—with a fundamentally different design than other previously published models. Instead of providing pixel-level cell segmentations within the tumour bed and explicitly following the TILs Working Group guideline, our model efficiently and quickly regresses the WSI-level TIL score in a single step using features extracted from the WSI through an open-source foundation model, without required annotations or explicit post-processing. Therefore, ECTIL does not require pixel-level segmentation or cell annotations, but only whole slide level TIL scores, to be trained. Based on previously generated evidence, this computational TIL assessment model builds upon an existing deep learning-based WSI-level TIL regression method with an improved foundation model, gated attention mechanism, and an additional projection head for feature adaptation.¹⁵ ECTIL can be trained with only a few hundred WSI-level TIL scores and is a straightforward, easy-to-train tool for accurately identifying and quantifying TILs on H&E-stained WSIs of either biopsies or full resections from various scanners of patients with breast cancer across molecular and morphological subtypes. We validate ECTIL on the largest known computational TIL assessment model validation dataset to date and compare it with an open-source, validated computational TIL assessment model that uses the traditional segmentation of cells. ECTIL has the potential to streamline and automate the TIL scoring process in breast cancer, facilitate the use of TIL scores as a biomarker in translational research, and ultimately enhance its integration into clinical practice. Furthermore, ECTIL will be integrated in a prospective de-escalation phase 3 clinical trial in patients with low stage triple-negative breast cancer (NCT06476119).

Methods

Study design and cohorts

The current study and data collection were approved by the Institutional Review Board of the Netherlands Cancer

Institute. All patients included in the clinical trials provided written informed consent for study treatment and use of their derived patient data for the purpose of other translational research. The paradigm initiative was approved by the ethical committee of the Netherlands Cancer Institute (NKI) and the NKI translational research board (CFMPB173). All included cohorts were conducted according to the Good Clinical Practice guidelines and the Declaration of Helsinki.

Individual clinical and histopathological data including WSIs from primary tumour material with an available pathologist TIL score were collected from six cohorts of patients with early breast cancer. We did not define eligibility criteria for the selected cohorts. Two cohorts consisted of publicly available data. The Cancer Genome Atlas (TCGA) is a large cancer genomics dataset that has molecularly characterised primary cancer samples of different cancer types, including breast cancer ($n=1134$). The clinical variables were exported from the Genomic Data Commons repository. The second cohort consisted of data acquired through the Breast Cancer Somatic Genetics Study (BASIS, $n=560$) as published by Nik-Zainal and colleagues.¹⁵ TCGA included patients treated in the USA and BASIS was a UK study of an international cohort.¹⁵ Three cohorts consisted of patients treated in randomised clinical trials, all conducted in the Netherlands (N4+,¹⁶ $n=855$; MATADOR,¹⁷ $n=664$; and TRAIN II,¹⁸ $n=438$). The inclusion flow diagram for the current study is depicted in appendix 1 (p 8). A detailed description of each trial design is given in appendix 1 (pp 2–3).

The PARADIGM cohort, which was used as an external test set only, comprised retrospectively collected clinical data and primary tumour material from patients younger than 40 years with triple-negative breast cancer (any T stage, node negative, and metastasis free) who had not received systemic therapy and were treated between 1989 and 2000 in the Netherlands, as part of the PARADIGM initiative.¹⁹ Date ranges for enrolment and follow-up of each cohort are shown in appendix 1 (pp 2–3).

The study design and model details were decided beforehand with no post-hoc changes, although an explicit protocol was not prepared. This study was not registered. There was no patient or public involvement during the design, conduct, reporting, interpretation, or dissemination of the study. We adhered to the TRIPOD+AI reporting guidelines (appendix 2).

Data extraction

For this current study, we received the required clinical variables directly from the principal investigator of the study. Tumour grade (according to the Bloom-Richardson classification),²⁰ presence of lymphovascular invasion, and breast cancer histology and subtype (hormone and HER2 receptor status) were assessed locally according to national guidelines,²¹ or centrally, according to the study protocol. Ethnicity data were not obtained due to privacy reasons in the Netherlands.

See Online for appendix 1

See Online for appendix 2

TILs were scored on H&E FFPE WSIs according to previously published recommendations by the TILs Working Group.^{7,22} All TIL scores were collected retrospectively. TCGA, BASIS, and PARADIGM were scored by a single pathologist (RS) and N4+ and MATADOR were scored by a pathologist (HMH) and by MO (supervised by HMH), because their scoring showed high interobserver concordance (appendix p 3). For TRAIN II the TIL scoring process has been described previously.²³ In short, each slide was scored by RS and by another independent pathologist (either HMH or MO). In case an absolute difference of more than 10% was observed, the slide was reviewed, and a final score was given by RS. In case an absolute difference of 10% or less was observed, the geometric means were calculated after adding 0.5 to values equal to 0. This averaging rule was used according to the previously published TRAIN II manuscript. Interobserver concordance was high between both RS and HMH and RS and MvS (appendix 1 p 3). There were cases in which two WSIs and only one TIL score were available for a patient. For TCGA, both slides were included with the same label for 14 patients after manual review showed that the slides were different, of high quality, and representative for the provided TIL score. For all other cohorts a single WSI with the highest quality was used. Quality was assessed by the appointed pathologist. TILs were scored on primary tumours that were removed during surgery before any neoadjuvant systemic therapy, except for the TRAIN II cohort, in which TILs were scored on the diagnostic biopsy.

WSIs were included in the current study if a scanned WSI and TIL score by a pathologist was already available and if the clinical variables corresponding to the patient were available, which in some cohorts reduced the number of patients that could be included in the current study (appendix 1 pp 8, 29). No further patient or WSI selection was made. WSIs were scanned by different scanners including Leica Biosystems Aperio (all samples of TCGA and 425 samples of N4+), Hamamatsu C9600-12 (BASIS), or 3DHistech P1000 (PARADIGM, MATADOR, TRAIN II, and 127 samples of N4+). For the TCGA cohort, 14 patients had two representative WSIs that were both included in the ECTIL models; a note about safeguards to prevent overfitting on these patients is provided in appendix 1 (pp 6–7). The WSI selection workflow is depicted in appendix 1 (p 8). In short, we collected 5799 WSIs over six cohorts. The pathologist's TIL scores were provided for 2480 patients. Complete clinical variables were available for 2340 patients with 2354 slides to perform concordance analyses. In the PARADIGM cohort, 476 WSIs corresponded to 390 patients. However, for the univariate and multivariate survival analyses in PARADIGM, 384 patients were included, because six patients had missing clinical data (five patients with missing T stage and one patient with no information on lymphovascular invasion; appendix 1 p 8). Cases with missing clinical data were excluded from the analysis.

Deep learning model: ECTIL

All steps of ECTIL are visualised in figure 1. To exclude non-tissue regions, background was detected and accordingly discarded using the Foreground Extraction from Structure Information algorithm, as implemented in the open-source Deep Learning Utilities for Pathology (version 0.3.22) library developed at the Netherlands Cancer Institute.^{24,25} Tiles with no resulting foreground pixels (ie, no tissue) were removed. Qualitative results of samples from each dataset are shown in appendix 1 (pp 11–19).

We used RetCCL, a publicly available convolutional neural network that was pre-trained with self-supervised learning on pathology slides by Wang and colleagues,²⁶ to extract 2048-dimensional feature vectors from non-overlapping 512×512 pixel patches at 0.5 µm per pixel, after normalisation using ImageNet statistics (figure 1).²⁷ RetCCL was chosen due to its superior performance shown by Niehues and colleagues²⁸ at the time ECTIL was trained. Because RetCCL was trained on only slides from TCGA, there is no possible data leakage from any of the external datasets we validated the model on throughout this paper.

We used a gated attention-based multiple instance learning model to directly regress TIL scores from the extracted WSI patch features, based on previous work.^{29,30} We added a non-linear layer with a rectified linear unit layer after the extracted feature vectors to generate a new task-specific tile encoding and a sigmoid layer to map the model output to the [0,1] range. Details of the hyperparameter search, the final parameter set, and the exact implementation of ECTIL's model architecture with these parameters are described in appendix 1 (pp 3–4); this configuration was used to train all models. We compared ECTIL with EasTIL, an open-source cell segmentation pipeline that was previously evaluated as a computational TIL assessment model (implementation described in appendix 1 pp 4–5).³¹ For this comparison, we required tumour bed annotations, which were collected for a subset of PARADIGM as described in appendix 1 (p 5).

ECTIL-TCGA was trained on the TCGA cohort (356 slides), using five-fold cross validation split by medical centre. The model was trained on three folds, validated during training on one fold, and internally tested on the remaining fold (figure 1). Finally, the model was externally tested on all other cohorts. ECTIL-combined was trained on all cohorts except PARADIGM (1964 slides) using a five-fold leave-one-cohort-out data split, with one cohort held out for validation during training in each fold, and was tested only on PARADIGM. ECTIL-triple-negative breast cancer (TNBC) was trained on only triple-negative breast cancer samples from TCGA, BASIS, N4+, and MATADOR (400 slides) using a four-fold leave-one-cohort-out data split, with one cohort held out for internal validation in each fold, and was tested only on PARADIGM. For external testing of each ECTIL model, we selected the model with the highest Pearson's *r* on the validation set of each fold, using the ensemble of models for the test set. EasTIL used a

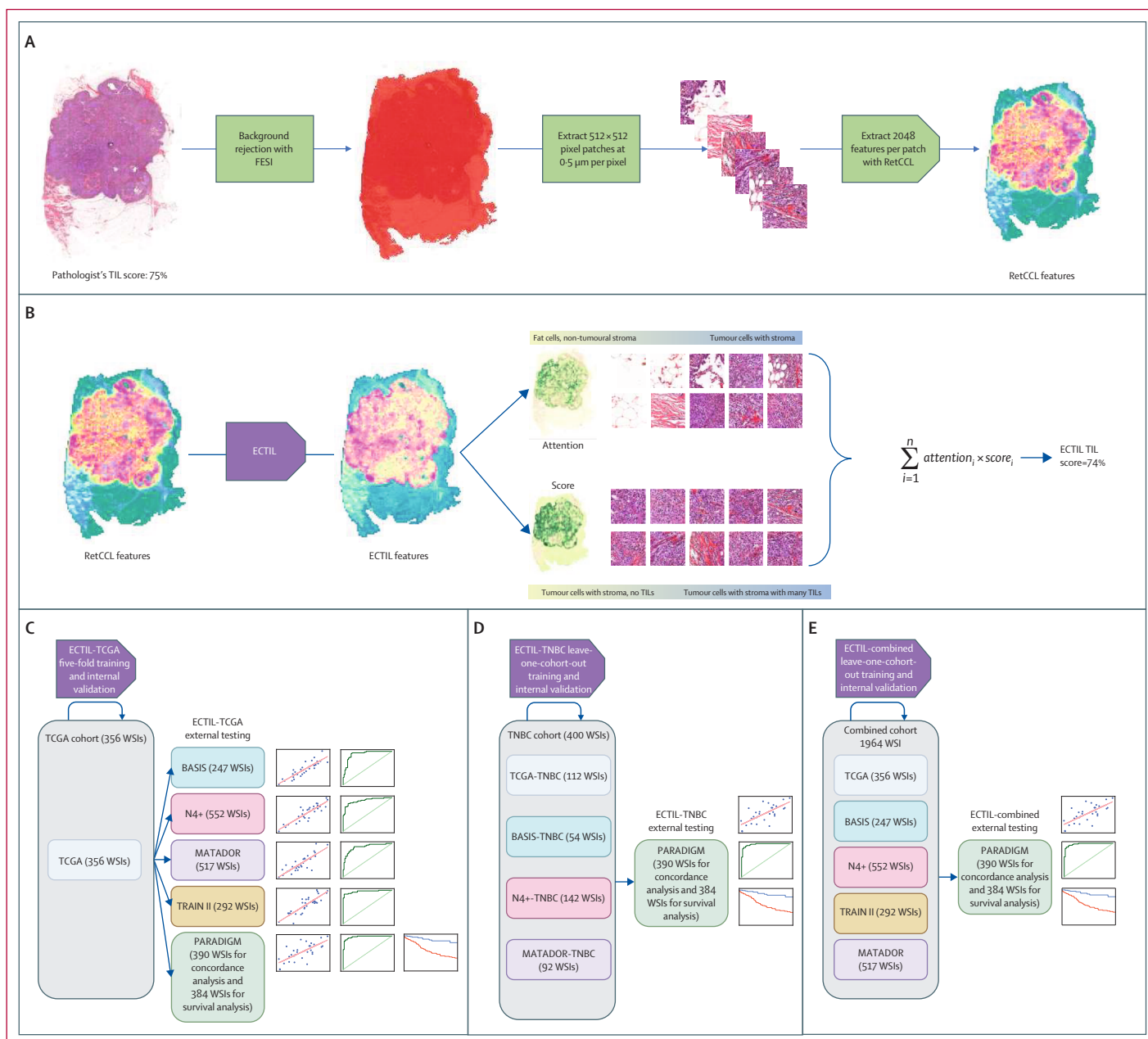


Figure 1: Schematic overview of data preprocessing, training, internal validation, external testing set-up, and the single-step inference process of the ECTIL model

(A) Data preprocessing with open-source tools. (B) Single-step TIL prediction with ECTIL. The single-step ECTIL inference took as input the RetCCL features of a WSI, with a ground-truth pathologist's TIL score of 75%, that were mapped to a 512-dimensional feature space, which was used to compute the attention and the score for each tile. For both the attention and the score, we showed the heatmap for this sample and picked random tiles over the score and attention distribution and describe the content of these patches. Furthermore, the attention-weighted mean score of all tiles was the final model prediction, which in this case was 74%, very close to the pathologist's score. (model details are found in appendix 1 p 3). (C) Training, internal validation, and external testing set-up for ECTIL-TCGA. (D) Training, internal validation, and external testing set-up for ECTIL-TNBC. (E) Training, internal validation, and external testing set-up for ECTIL-combined. The scatter plots, ROC curves, and survival curves were generated with synthetic data for illustration purposes only. ECTIL=label-efficient computational stromal TIL assessment. FESI=Foreground Extraction from Structure Information. TCGA=The Cancer Genome Atlas. TIL=tumour-infiltrating lymphocyte. TNBC=triple-negative breast cancer. WSI=whole slide image.

pre-trained cell segmentation network and was implemented using open-source model weights.

Statistical analysis

We performed concordance analyses by computing the mean squared error, Pearson correlation coefficient, Spearman's rank correlation coefficient, and concordance

correlation coefficient in Python using torchmetrics.³² To analyse the model's capacity to identify high versus low TILs we split the pathologist score into TILs-low (<30%) and TILs-high (≥30%), although a universal cutoff score is yet to be established. Using these binary ground-truth labels and the model's predicted TIL score, we computed the area under the receiver operating characteristic curve

(AUROC). We also calculated the average precision and AUROC for varying cutoff points (10%, 50%, and 75%; appendix 1 pp 33–34). We visualised model predictions over datasets and slides using scatter plots, calibration curves, and heatmaps of score and attention.

We reproduced the overall survival analysis as reported by de Jong and colleagues⁵ in R using the *survival* package on the PARADIGM cohort. To test the effect of the pathologist's and ECTIL's TIL scores on overall survival we calculated hazard ratios (HRs) and concordance indexes using univariate and multivariate Cox proportional hazard models, with adjustment for other clinicopathological characteristics (including T stage, tumour grade, histology, lymphovascular invasion, and local treatment) in the PARADIGM cohort. The ECTIL and EasTIL scores were normalised to the minimum and maximum of the training dataset. We produced Kaplan–Meier plots for the pathologist's score with previously defined cutoffs (30% and 75%), because these cutoffs previously showed the prognostic value of TILs in young (aged <40 years) patients with node-negative triple-negative breast cancer.⁵ The statistical significance was calculated using the log-rank test as implemented in the *ggsurvplot* package in R. To compare the pathologist's score with that of ECTIL we generated Kaplan–Meier plots, dichotomising both scores at their respective medians and evaluating differences in survival between the resulting groups. The Schoenfeld residuals were used to test the proportionality of hazards assumption over time.

Role of the funding source

The funders of the study had no role in the study design, data collection, data analysis, data interpretation, or writing of the report.

Results

The baseline characteristics of all collected cohorts are set out in table 1, with a total of 2340 patients with an available scanned WSI and complete clinical data. The median age across the cohorts excluding PARADIGM (ECTIL-combined column; appendix 1 p 30) was 49 years (IQR 43–56). Excluding the 390 patients from PARADIGM, 1943 (99%) of 1964 patients were female. Most patients had stage II (842 [43%] of 1964 patients) or stage III (864 [44%] of 1964 patients) disease at time of diagnosis or study entry. 12 (1%) of 1964 patients had known stage IV disease at study entry. Most tumours (1308 [67%] of 1964 patients) were categorised as breast cancer of no special type, with either tumour grade II (655 [33%] of 1964 patients) or III (675 [34%] of 1964 patients). Presence or absence of lymphovascular invasion was unknown for 1742 (89%) of 1964 tumours. 1175 (60%) of 1964 tumours were oestrogen-receptor positive and HER2 negative (ER+ and HER2–), 230 (12%) of 1964 tumours were ER+ and HER2+, 123 (6%) of 1964 tumours were ER– and HER2+, and 400 (20%) of 1964 tumours were triple-negative breast cancer. The median pathologist TIL score was 10% (IQR 5–25).

For the PARADIGM cohort (table 1), the median age was 35 years (IQR 32–38). All patients were female and premenopausal. Stage I disease was most prevalent (232 [59%] of 390 patients), followed by stage II (157 [40%] of 390 patients). All patients were node negative at diagnosis, with no known distant metastases. Most tumours were invasive breast cancer of no special type (333 [85%] of 390 tumours). 20 (5%) of 390 patients had metaplastic carcinoma. 336 (86%) of 390 tumours were grade 3 and 52 (13%) of 390 were grade 2. Lymphovascular invasion was present in 40 (10%) of 390 tumours. All tumours in PARADIGM were triple-negative breast cancer. The median pathologist TIL score was 20% (IQR 5–70).

An overview of the baseline characteristics of the ECTIL-TNBC and ECTIL-combined cohorts, which are used for training ECTIL-TNBC and ECTIL-combined models, is provided in appendix 1 (p 30). Figure 1 summarises the dataset sizes and compositions that ECTIL-TGCA, ECTIL-TNBC, and ECTIL-combined were trained on.

First, we compared the predictions of ECTIL-TGCA with the pathologist scores in the TCGA internal test set, BASIS, N4+, TRAIN II, MATADOR, and PARADIGM cohorts. When evaluated on the internal test set of the TCGA cohort, ECTIL-TGCA attained a Pearson correlation coefficient (r) of 0.61 and an AUROC of 0.84 (appendix 1 p 33). ECTIL-TGCA produced, across five external test cohorts, an r of 0.54–0.74 and an AUROC of 0.80–0.94 (figure 2). ECTIL showed the highest concordance on MATADOR (r 0.74; AUROC 0.94) and the lowest on PARADIGM (r 0.58; AUROC 0.80) or N4+ (r 0.54; AUROC 0.82), depending on the metric of interest. Additional metrics and the AUROC for varying pathologist score cutoffs (10%, 30%, and 75%) can be found in the appendix (pp 33–34). Case-level predictions are presented through scatter plots (appendix 1 p 9). A thorough subset analysis presents the model performance within and across datasets grouped by all clinical variables and scanner types (appendix 1 p 35).

Second, we evaluated ECTIL-TNBC and ECTIL-combined on the PARADIGM cohort. ECTIL-TNBC showed a higher performance (r 0.64; AUROC 0.83) than ECTIL-TGCA (r 0.58; AUROC 0.80; figure 2). ECTIL-combined, which was trained on all slides from all cohorts except PARADIGM ($n=1964$), showed an r of 0.69 and an AUROC of 0.85 (figure 2). Additional metrics and the AUROC for varying pathologist score cutoffs can be found in the appendix (p 34). On the PARADIGM cohort, predictions from both ECTIL-TNBC and ECTIL-combined were predictive of the pathologist's TIL scores binarised at thresholds of 10%, 30%, 50%, and 75%, with AUROC values of 0.86, 0.85, 0.85, and 0.90 for ECTIL-combined and 0.82, 0.83, 0.82, and 0.88 for ECTIL-TNBC, respectively.

To further investigate the predictions of ECTIL-combined on the PARADIGM cohort, we constructed a calibration plot in figure 3 (calibration plots for ECTIL-TGCA and ECTIL-TNBC on the PARADIGM cohort are in appendix 1 p 10). We plotted the distribution of pathologist scores (y axis) against the predicted scores of ECTIL-combined,

	TCGA (N=356*)	BASIS (N=247)	N4+ (N=552)	MATADOR (N=517)	TRAIN II (N=292)	PARADIGM (N=390)
Age, years						
Median (IQR)	50·0 (43·0–62·0)	57·0 (48·0–69·0)	46·0 (40·0–49·0)	51·0 (46·0–57·0)	49·0 (43·0–57·0)	35·0 (32·0–38·0)
Missing	14 (4%)*	0	0	0	0	0
Gender						
Female	338 (95%)	244 (99%)	552 (100%)	517 (100%)	292 (100%)	390 (100%)
Male	4 (1%)	3 (1%)	0	0	0	0
Missing	14 (4%)*	0	0	0	0	0
Menopausal status						
Premenopausal	0	58 (23%)	0	266 (51%)	160 (55%)	390 (100%)
Perimenopausal	0	4 (2%)	0	0	20 (7%)	0
Postmenopausal	0	128 (52%)	0	245 (47%)	110 (38%)	0
Missing	356 (100%)*	57 (23%)	552 (100%)	6 (1%)	2 (1%)	0
Disease stage						
I	44 (12%)	41 (17%)	0	55 (11%)	0	232 (59%)
II	204 (57%)	85 (34%)	0	352 (68%)	201 (69%)	157 (40%)
III	71 (20%)	41 (17%)	552 (100%)	109 (21%)	91 (31%)	0
IV	10 (3%)	2 (1%)	0	0	0	0
Missing	27 (8%)*	78 (32%)	0	1 (<1%)	0	1 (<1%)
Primary tumour size (T stage)						
No evidence of primary tumour (T0)†	0	0	0	0	1† (<1%)	0
Tumour ≤2 cm (T1)	0	59 (24%)	129 (23%)	236 (46%)	17 (6%)	232 (59%)
Tumour >2 cm but ≤5 cm (T2)	0	92 (37%)	328 (59%)	249 (48%)	186 (64%)	151 (39%)
Tumour >5 cm (T3)	0	20 (8%)	85 (15%)	29 (6%)	75 (26%)	6 (2%)
Tumour with chest wall or skin involvement (T4)	0	8 (3%)	0	2 (<1%)	12 (4%)	0
Cannot be assessed (TX)	0	68 (28%)	0	0	1 (<1%)	0
Missing	356 (100%)*	0	10 (2%)	1 (<1%)	0	1 (<1%)
Regional lymph node involvement (N stage)						
No lymph nodes involved (N0)	0	83 (34%)	0	107 (21%)	114 (39%)	390 (100%)
1–3 lymph nodes involved (N1)	0	58 (23%)	0	315 (61%)	138 (47%)	0
4–9 lymph nodes involved (N2)	0	25 (10%)	0	74 (14%)	21 (7%)	0
≥10 lymph nodes involved (N3)	0	4 (2%)	0	21 (4%)	19 (7%)	0
Cannot be assessed (NX)	0	77 (31%)	0	0	0	0
Missing	356 (100%)*	0	552 (100%)	0	0	0
Distant metastasis (M stage)						
No distant metastasis (M0)	0	73 (30%)	552 (100%)	517 (100%)	290 (99%)	390 (100%)
Distant metastasis present (M1)	0	2 (1%)	0	0	0	0
Cannot be assessed (MX)	0	172 (70%)	0	0	2 (1%)	0
Missing	356 (100%)*	0	0	0	0	0
Tumour grade						
1	0	32 (13%)	121 (22%)	53 (10%)	12 (4%)	2 (1%)
2	0	90 (36%)	207 (38%)	220 (43%)	138 (47%)	52 (13%)
3	0	124 (50%)	205 (37%)	221 (43%)	125 (43%)	336 (86%)
Missing	356 (100%)*	1 (<1%)	19 (3%)	23 (4%)	17 (6%)	0
Lymphovascular invasion						
Absent	0	182 (74%)	0	0	0	349 (89%)
Present	0	40 (16%)	0	0	0	40 (10%)
Missing	356 (100%)*	25 (10%)	552 (100%)	517 (100%)	292 (100%)	1 (<1%)
Breast cancer subtype						
ER+ and HER2+	32 (9%)	3 (1%)	0	8 (2%)	187 (64%)	0
ER+ and HER2–	168 (47%)	188 (76%)	410 (74%)	409 (79%)	0	0
ER– and HER2+	9 (3%)	2 (1%)	0	7 (1%)	105 (36%)	0
Triple negative	112 (31%)	54 (22%)	142 (26%)	92 (18%)	0	390 (100%)
Missing	35 (10%)*	0	0	1 (<1%)	0	0

(Table 1 continues on next page)

	TCGA (N=356*)	BASIS (N=247)	N4+ (N=552)	MATADOR (N=517)	TRAIN II (N=292)	PARADIGM (N=390)
(Continued from previous page)						
Histology						
Breast cancer of no special type	321 (90%)	193 (78%)	115 (21%)	414 (80%)	265 (91%)	333 (85%)
Invasive lobular carcinoma	18 (5%)	17 (7%)	50 (9%)	71 (14%)	13 (4%)	1 (<1%)
Metaplastic	0	3 (1%)	0	0	0	20 (5%)
Other	3 (1%)	34 (14%)	39 (7%)	25 (5%)	14 (5%)	36 (9%)
Missing	14 (4%)*	0	348 (63%)	7 (1%)	0	0
Pathologist TIL score (%)						
Median (IQR)	12.5 (5–30)	13.0 (8–30)	15.0 (10–30)	10.0 (5–20)	5.00 (2–20)	20.0 (5–70)
Missing	14 (4%)*	0	0	0	0	0

Data are n (%), unless otherwise stated. ER=oestrogen receptor. HER2=human epidermal growth factor receptor 2. TCGA=The Cancer Genome Atlas. TIL=tumour-infiltrating lymphocyte. *In TCGA, 14 patients contributed two slides each, resulting in 356 total samples from 342 unique patients. The table presents patient-level clinical characteristics, so the additional 14 slides (duplicates) do not have separate clinical data. †Patients with no evidence of a tumour in the breast but cancer has spread to axillary lymph nodes.

Table 1: Baseline characteristics of all clinical cohorts

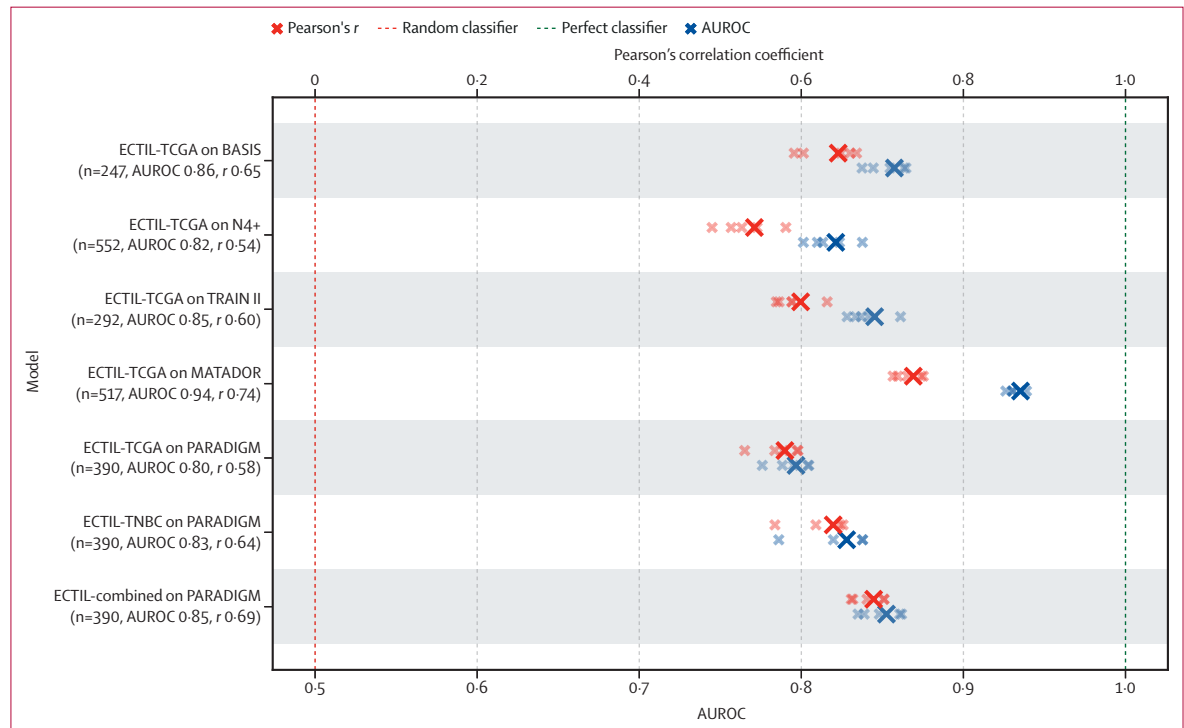


Figure 2: Overview of performance of all models on all datasets

The large cross marks represent the metric for the ensemble that was used for further analysis. The transparent, small cross marks show the results of each separate model that has been trained on various folds, to show how sensitive the predictions are to various training data. The bottom x axis displays the AUROC, for which the pathologist score is split at a TIL score of 30 to generate high TILs and low TILs labels. The top x axis displays the Pearson's correlation coefficient (r). The y axis displays the test results of various models on various cohorts. A random model would attain a Pearson's correlation coefficient of 0 and an AUROC of 0.5, whereas a perfect model would attain a Pearson's correlation coefficient of 1.0 and AUROC of 1.0. N refers to the number of samples of the dataset that we computed the metric for. AUROC=area under the receiver operating characteristic curve. ECTIL=label-efficient computational stromal TIL assessment. TCGA=The Cancer Genome Atlas. TIL=tumour-infiltrating lymphocyte. TNBC=triple-negative breast cancer.

binned in intervals of 0.05 (x axis). The distribution of ECTIL-combined scores was smaller (0.0–0.7) than the pathologist scores (0.1–0.95). Moreover, ECTIL-combined tended to overpredict the samples with a low (<0.2) pathologist score and underpredict samples with a higher (>0.3) pathologist score. We also observed discordant cases: for each sample with a pathologist score of 0.9, ECTIL could

predict these to have a value of 0.2–0.7, whereas samples that were predicted to have a value of 0.2–0.5 could correspond to samples originally scored by a pathologist to have a TIL score of 0 or 0.9.

On the annotated subset of PARADIGM (dataset description in appendix 1 p 5 and performance metrics in appendix 1 p 43), ECTIL-combined and the open-source cell

segmentation, detection, and post-processing TIL calculation pipeline EasTIL obtain an r of 0.77 and 0.53 and an AUROC of 0.88 and 0.78, respectively.

Next, we qualitatively analysed samples with large discrepancies between ECTIL and pathologist scores to better understand ECTIL's limitations. For each cohort and for two groups (pathologist TIL scores <30% and >50%, which we term pTIL-l and pTIL-h, respectively) we randomly chose one sample from the top ten samples with the largest differences between the predicted score and the pathologist score. For these samples, we visualised the heatmap of ECTIL-predicted scores and attention weights for each patch, which were analysed by a board-certified pathologist. These samples are presented in appendix 1 (pp 11–18). The first limitation was on samples from invasive lobular carcinoma, for which tumour cells were scored as TILs. Additionally, patches that contained lymphovascular invasion, other immune hotspots, fibrotic areas, and lymphocytes at the tumour boundary received a high TIL score and attention score, whereas a pathologist would not include these lymphocytes. Cases with large WSIs and small tumour beds could receive a different score due to the accumulated attention weight of the large number of patches with low attention. In the aggregated results of the subset analysis presented in appendix 1 (p 35), we find that ECTIL-TCGA indeed performed quantitatively worse on patients with invasive lobular carcinoma (r 0.36, AUROC 0.79) across datasets than on those with, for example, breast cancer of no special type (r 0.58; AUROC 0.83). For lymphovascular invasion, however, we do not find a systematic performance decrease over the aggregated results. The Pearson correlation coefficient was slightly higher for presence (r 0.66) than for absence (0.6) of lymphovascular invasion, whereas the AUROC was slightly lower for the presence (AUROC 0.79) compared with absence (0.83) of lymphovascular invasion.

We assessed whether the predicted TIL scores by ECTIL-TCGA, ECTIL-TNBC, and ECTIL-combined were independently associated with overall survival. In the univariate Cox regression analysis with TILs in 10% increments, all three ECTIL TIL scoring models yielded statistically significant HRs (ECTIL-TCGA 0.87, 95% CI 0.79–0.96; $p=0.0010$; ECTIL-combined 0.86, 0.78–0.94; $p=0.0002$; ECTIL-TNBC 0.85, 0.77–0.94; $p=0.0002$), similar to the pathologist TIL scores (HR 0.87, 95% CI 0.82–0.92; $p<0.0001$; table 2). In the multivariate Cox regression the HR of 10% TIL increments of all ECTIL models (ECTIL-TCGA HR 0.87, 95% CI 0.79–0.95; $p=0.0032$; ECTIL-combined 0.85, 0.77–0.93; $p=0.0007$; ECTIL-TNBC 0.84, 0.76–0.93; $p=0.0007$) remained significant, independent of all other clinical characteristics, and similar to the pathologist score (HR 0.86, 95% CI 0.81–0.92; $p<0.0001$; table 2).

Cox regression results for all ECTIL models and EasTIL on the annotated subset of PARADIGM are provided in appendix 1 (p 44). On all tissue, ECTIL-combined yielded an HR of 0.81 (95% CI 0.69–0.95; $p=0.0078$). EasTIL, on

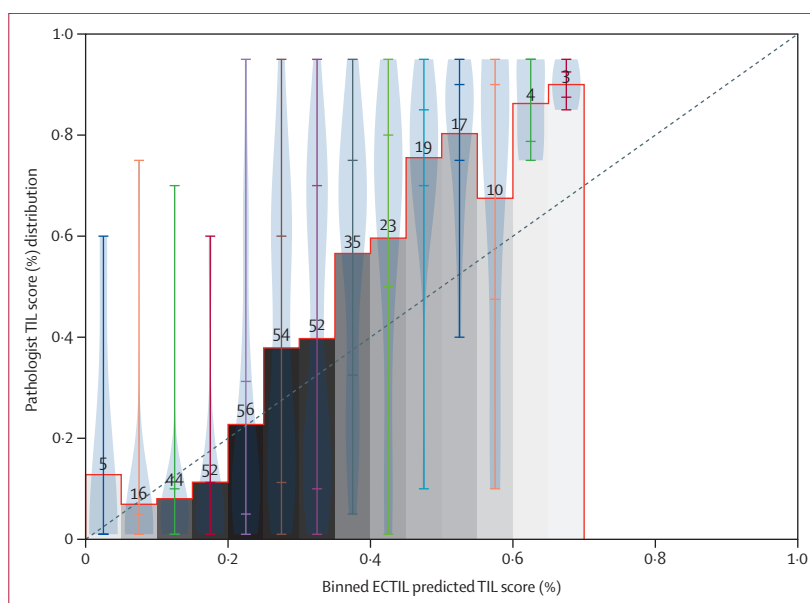


Figure 3: Calibration curve of ECTIL-combined on PARADIGM

This figure displays a calibration curve of the ECTIL model trained on all cohorts and validated on the PARADIGM cohort (ECTIL-combined). The curve shows the distribution of pathologist scores conditional on the binned model-predicted scores $p(y|\hat{y})$, where y is the ground-truth score, and $\hat{y}$ denotes the predicted score bin. A perfectly predictive and calibrated model would place the bars exactly along the $x=y$ line across the entire distribution. The x axis represents 20 uniform bins of ECTIL predictions, whereas the y axis shows the distribution of pathologist's scores corresponding to the samples in that bin. The colour of each bar represents the number of samples in the bin (the bin with the most samples is black and the bin with the least samples is white), and the exact numbers are displayed above the bar. For each bin, violin plots show the distribution of pathologist scores (minimum, 10th percentile, 90th percentile, and maximum), whereas the bar plot displays the mean score for that bin. ECTIL=label-efficient computational stromal TIL assessment. TIL=tumour-infiltrating lymphocyte.

tumour bed tissue, attained an HR of 0.81 (0.61–1.07; $p=0.14$). When ECTIL-combined was run only on the annotated tumour bed tissue, it maintained statistical significance (HR 0.83, 95% CI 0.71–0.97; $p=0.0018$).

We generated Kaplan–Meier curves for the PARADIGM cohort using multiple scoring methods. First, we applied pathologist TIL score cutoffs previously used in the literature (<30%, 30%–74%, and $\geq 75\%$; appendix 1 p 20). Second, we assessed pathologist TIL scores using a median cutoff (≥ 20 vs <20; appendix 1 p 21). Third, we analysed ECTIL-combined scores, dichotomised at the median (≥ 27 vs <27; appendix 1 p 22). The analysis revealed significant differences in overall survival between the groups. Patients with pathologist TIL scores $\geq 75\%$ showed a 3-year overall survival of 100% (95% CI 100.0–100.0), compared with 89% (82.9–95.3) for pathologist TIL scores of 30–74%, and 81% (75.4–86.2) for scores <30% ($p<0.0001$). With the median cutoff, pathologist TIL scores $\geq 20\%$ corresponded to 94% (95% CI 90.6–97.5) 3-year overall survival, whereas scores <20% corresponded to 80% (74.8–85.8) 3-year overall survival ($p<0.0001$). For ECTIL-combined TIL scores, a median cutoff ≥ 27 corresponded to 90% (95% CI 85.8–94.4) 3-year overall survival, compared with 83% (78.3–88.8) 3-year overall survival for scores <27 ($p<0.0043$). These findings highlight the clinical value of both ECTIL and pathologist scores when split by the median

	No TIL score		Pathologist TIL score		ECTIL-TCGA TIL score		ECTIL-combined TIL score		ECTIL-TNBC TIL score	
	HR (95% CI)	p value	HR (95% CI)	p value	HR (95% CI)	p value	HR (95% CI)	p value	HR (95% CI)	p value
Univariate analysis*										
TILs (10% increment)	..	..	0.87 (0.82–0.92)	<0.0001	0.87 (0.79–0.96)	0.0038	0.86 (0.78–0.94)	0.0012	0.85 (0.77–0.94)	0.0013
Multivariate analysis†										
TILs (10% increment)	..	..	0.86 (0.81–0.92)	<0.0001	0.87 (0.79–0.95)	0.0032	0.85 (0.77–0.93)	0.0007	0.84 (0.76–0.93)	0.0007
T stage										
T1a/b	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..
T1c	1.02 (0.55–1.91)	0.95	1.02 (0.55–1.91)	0.95	1.11 (0.59–2.09)	0.74	1.09 (0.58–2.05)	0.79	1.11 (0.59–2.09)	0.74
T2/3	1.50 (0.81–2.80)	0.20	1.62 (0.87–3.02)	0.13	1.62 (0.87–3.02)	0.13	1.62 (0.87–3.02)	0.13	1.66 (0.89–3.10)	0.11
Tumour grade										
Grade 1–2	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..
Grade 3	0.82 (0.52–1.28)	0.38	1.01 (0.64–1.60)	0.96	0.88 (0.56–1.39)	0.58	0.92 (0.58–1.46)	0.73	0.90 (0.57–1.43)	0.66
Histology										
Breast cancer of no special interest	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..
Metaplastic	0.34 (0.11–1.08)	0.068	0.25 (0.08–0.81)	0.020	0.32 (0.10–1.03)	0.055	0.32 (0.10–1.03)	0.056	0.32 (0.10–1.02)	0.055
Other	0.61 (0.32–1.16)	0.13	0.62 (0.32–1.20)	0.16	0.60 (0.31–1.15)	0.13	0.62 (0.32–1.18)	0.15	0.62 (0.32–1.18)	0.15
Lymphovascular invasion										
Absent	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..
Present	2.05 (1.30–3.23)	0.0021	1.87 (1.18–2.97)	0.0076	1.98 (1.25–3.14)	0.0036	2.05 (1.29–3.24)	0.0022	2.00 (1.26–3.17)	0.0031
Local treatment										
Breast conserving surgery plus radiotherapy	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..	1 (ref)	..
Mastectomy	1.39 (0.96–2.00)	0.16	1.41 (0.98–2.04)	0.067	1.47 (1.02–2.12)	0.041	1.47 (1.02–2.13)	0.038	1.48 (1.03–2.14)	0.035
Other	1.91 (0.96–2.00)	0.080	1.76 (0.71–4.36)	0.22	1.81 (0.73–4.49)	0.20	1.93 (0.78–4.79)	0.16	1.85 (0.75–4.60)	0.18

Univariate and multivariate Cox regression results, comparing the pathologist's score with the various ECTIL models that were trained on different subsets of data (ECTIL-TCGA, ECTIL-combined, and ECTIL-TNBC). Events refer to deaths. For both the univariate and multivariate regression models, all computational models attained a similar HR to the pathologist's score. ECTIL=label-efficient computational stromal TIL assessment. HR=hazard ratio. TILs=tumour-infiltrating lymphocytes. TCGA=The Cancer Genome Atlas. TIL=tumour-infiltrating lymphocyte. TNBC=triple-negative breast cancer. *Concordance indices were 0.62 (p<0.0001) for pathologist TIL score, 0.57 (p=0.0010) for ECTIL-TCGA TIL score, 0.58 (p=0.00018) for ECTIL-combined TIL score, and 0.58 (p=0.00017) for ECTIL-TNBC TIL score. †Concordance indices were 0.63 (p<0.0001) for no TIL score, 0.68 (p<0.0001) for pathologist TIL score, 0.65 (p<0.0001) for ECTIL-TCGA TIL score, 0.65 (p<0.0001) for ECTIL-combined TIL score, and 0.65 (p<0.0001) for ECTIL-TNBC TIL score.

Table 2: Cox regression results (N=384; events=139)

in patients younger than 40 years with systemic, treatment-naïve, low stage, node-negative, triple-negative breast cancer. The HRs for these groups further supported these findings, indicating the validity of these scoring methods in predicting overall survival (table 2; appendix 1 pp 20–22).

The Kaplan–Meier curves on the annotated PARADIGM subset for ECTIL-combined on all tissue, ECTIL-combined on tumour bed tissue, and EasTIL on tumour bed tissue are presented in appendix 1 (pp 23–25). Both ECTIL-combined approaches showed statistically significant stratification of overall survival: for ECTIL-combined run on all tissue, patients with scores ≥ 26 had a 3-year overall survival of 92% (95% CI 86.4–97.4) compared with 85% (78.4–93.1) for scores < 26 (p=0.0084); for ECTIL-combined run on only tumour bed tissue, patients with scores ≥ 27 had a 3-year overall survival of 91% (85.9–97.3) compared with 86% (79.3–93.4) for scores < 27 (p=0.0071). EasTIL run on only tumour bed tissue obtained a split that was not considered statistically significant (TIL score ≥ 1 corresponded to 90% [95% CI 84.3–95.7] 3-year overall survival, compared with 87% [80.1–94.9] for scores < 1 ; p=0.20).

Discussion

In this study, we present ECTIL, a label-efficient computational TIL assessment model that regresses the TIL scores

from extracted features of WSI patches directly. To our knowledge, this is the largest retrospective validation study of a computational TIL assessment model thus far, including 2340 patients with available breast carcinoma WSIs, pathologist-derived TIL scores, and clinical data, from six different cohorts. We showed that ECTIL, when trained on a limited dataset of only 200 WSIs, each labelled with a single, slide-level TIL score provided by one pathologist (eg, 30%), could produce scores that were highly concordant with a pathologist's assessments across multiple external test cohorts. This concordance was reflected in performance metrics, including Pearson correlation coefficient and AUROC values, which indicated strong agreement with pathologist evaluations. Additionally, ECTIL-generated TIL scores showed a significant association with overall survival in patients younger than 40 years with node-negative triple-negative breast cancer independent of other clinicopathological variables, suggesting the clinical utility of ECTIL for distinguishing patients with triple-negative breast cancer at lower risk of dying without therapy who could forgo adjuvant or neoadjuvant chemotherapy. Our findings revealed that expanding the training dataset to include larger and more heterogeneous cohorts enhanced ECTIL's concordance with pathologist assessments. Although ECTIL-TCGA and ECTIL-combined were

trained on WSIs from mostly patients with ER+ breast cancer (200 [56%] of 356 WSIs for ECTIL-TGCA and 1405 [72%] of 1964 WSIs for ECTIL-combined), they showed considerable concordance with the pathologist when evaluated on a dataset of patients younger than 40 years with node-negative triple-negative breast cancer. These results underscore the importance of training ECTIL on diverse datasets to maximise model performance. Although the obtained Pearson correlation coefficients did not suggest a perfect performance, the inter-rater and intra-rater variability of pathologists sets an upper limit to the obtainable performance. For example, in the MATADOR dataset, the correlation coefficient between two pathologists was 0.8, whereas ECTIL-TGCA had a coefficient of 0.74 with the mean of the pathologists' scores. This finding indicates that the concordance between ECTIL-TGCA and the mean of the pathologists was nearly as strong as the concordance observed between two pathologists. This generalisability is also observed across scanner types. ECTIL-TGCA, which was only trained on slides scanned on an Aperio scanner, did not consistently underperform on slides from other scanners across all datasets (appendix 1 p 35). Within the N4+ dataset, however, ECTIL-TGCA performed better on slides scanned on an Aperio scanner than on those scanned on the 3DHitech scanner, suggesting that the model is not perfectly invariant to scanner type. We hypothesise that this is one of the reasons that ECTIL-combined performed better on the PARADIGM dataset, because it had been additionally trained on 3DHitech slides, the same scanner used for PARADIGM. Our choice for using PARADIGM as an exclusive test set for ECTIL was that it had previously shown TILs to be a clear prognostic biomarker in patients younger than 40 years with triple-negative breast cancer who were not treated with chemotherapy, which has direct consequences for the treatment of these patients.⁵ However, to understand the performance of ECTIL scores for the prediction of overall survival in other breast cancer subtypes, separate validation studies should be performed. Our results suggest that ECTIL scores are concordant with pathologist scores over molecular subtypes.

In contrast to existing models that generally use a detection–segmentation pipeline involving multiple models and post-processing steps, ECTIL is a single model trained on two orders of magnitude fewer annotations that finishes training from scratch in only 10 min on a commodity graphics processing unit (NVIDIA GeForce RTX 2080 Ti; appendix 1 p 5).^{10–13} Despite this fundamentally simple design, ECTIL showed a concordance with the pathologist that was highly similar to existing methods. For example, Bai and colleagues¹¹ obtained a Spearman's rank correlation coefficient (*r*) of 0.61 and 0.63 on two triple-negative breast cancer cohorts, whereas we found a Spearman's *r* of 0.47 to 0.72 across five cohorts with varying molecular subtype distributions, and 0.61 for our triple-negative breast cancer cohort. When representing the TIL scores as a fraction (from 0 to 1), Sun and colleagues¹⁰ reported a root mean squared error (RMSE) of 0.15 to

0.22 across three models and two triple-negative breast cancer datasets in their supplementary materials, and ECTIL obtained an RMSE of 0.14 to 0.28 across more heterogeneous cohorts. Thagaard and colleagues¹⁴ presented the results of commercial software with a Spearman's *r* of 0.79, which was much higher than other published scores. This model was developed using annotations guided by immunohistochemistry on slides that were picked as a held-out training set from the cohort collected at their institution. Hence, the generalisability of this model to external cohorts cannot be compared easily. Furthermore, Choi and colleagues¹³ presented the results of commercial software that reached a concordance correlation coefficient of 0.71 and 0.75 on two triple-negative breast cancer datasets, which were higher than most coefficients attained by ECTIL. Choi and colleagues' results have been obtained with a model trained on a large amount of training data; 2712 WSIs from multiple cancer types with a total of 2334 mm² dense tissue type annotations with 558 225 explicit cell annotations were used. When calibrating ECTIL scores (appendix 1 p 6), however, ECTIL aligned very well with the absolute pathologist TIL scores, and showed a concordance correlation coefficient of 0.67, closing the gap to the results presented by Choi and colleagues. This insight is particularly relevant given that many existing computational tumour analysis models have traditionally relied on more homogeneous training sets. Our study advocates for the inclusion of diverse data in training regimens to enhance model performance and predictive accuracy. These results suggest that ECTIL was able to generalise and perform well across different cohorts and could be suitable for implementation into daily clinical practice. Although a perfect comparison is not feasible, only commercial computational TIL assessment methods with extensive annotations appear to outperform ECTIL. Given that ECTIL was trained on substantially fewer annotations, it effectively addresses the main challenge highlighted by Thagaard and colleagues,¹⁴ which is the costly collection of annotations across diverse data distributions to develop generalisable algorithms.

Despite not explicitly following TILs Working Group guidelines, ECTIL learns to predict TIL scores through supervised training with guideline-based scores, suggesting that direct translation of these guidelines into computational frameworks might not be necessary. ECTIL's validation across numerous cohorts with diverse patient characteristics, technical preparations, scanners, and extraction strategies shows its generalisability. Through qualitative analysis of highly discordant cases by a board-certified pathologist who is trained to score TILs, we identified the limitations of ECTIL. The patch-level TIL scores matched the pathologist's idea of the number of stromal lymphocytes, and the attention heatmaps often showed a clear focus on mostly relevant tissue. Most discordant cases were caused by erroneously included areas such as lymphocytic hotspots, lymphocytes at the tumour border, lymphocytes in necrotic areas, and lymphocytes in

lymphovascular invasion. These issues were exacerbated when the tumour area was small relative to the WSI because there was more incorrect tissue to include. However, the TRAIN II cohort, which only included biopsies that had a low tumour-to-tissue ratio, did not lead to a significantly lower performance than ECTIL shows on resections. Similarly, the performance difference of ECTIL-combined folds that included or excluded training on biopsies from TRAIN II did not alter performance on PARADIGM. These results suggest that ECTIL's performance is invariant to extraction strategy of tissue. When running inference on only tumour bed annotations, the performance of ECTIL-combined did not improve further across most metrics, although the absolute scores did move closer to the pathologist scores (appendix 1 p 43). Another clear challenge was scoring invasive lobular carcinomas, which is a difficult task even for pathologists because the tumour cells have a similar morphology to lymphocytes and are potentially scored as such. These pitfalls (invasive lobular carcinomas and erroneous tissue inclusion) are highly similar to those in previous works as described by Thagaard and colleagues, although our model design is fundamentally different.¹⁴ ECTIL correctly identified pen markings and air bubbles, which receive low attention and were not the cause of discordance in the discordant cases we analysed, suggesting such artifacts are not an issue for ECTIL. A limitation that is unique to the architecture of ECTIL, however, is the dilution of scores in large WSIs with a small tumour bed. Due to the cumulative effect of many tiles with a low attention score, the scores of these tiles could still contribute 50% of the total weight. This dilution is the main reason our model predicted a maximum score of 70%, which could be solved using post-hoc calibration (appendix 1 p 6), tumour bed annotations (appendix 1 p 43), or by using a context-aware model architecture.

Our study had several limitations. We were limited to scanned WSIs with an available TIL score and complete clinical data. Although we did not observe any particular clinical imbalances of the excluded WSIs, we cannot completely rule out any unintentional selection bias. For example, there could have been possible selection bias based on ethnicity, because we were not able to obtain these data of the included patients. Patients without an available TIL score (eg, because of the quality of the WSI or if there was no tumour present) were not included in our study. However, by carefully tabulating the patient demographics of the included WSIs used for training and validation of the ECTIL-models, we hope to be transparent about the patient selection included in the current study. Another possible limitation of this study is that multiple pathologists were involved in the TIL scoring of the various studies, and this might have influenced the results of the model. However, all pathologists scored TILs according to the TIL guidelines and inter-observer agreement between the pathologists was good. Furthermore, only a single foundation model has been evaluated in this study, even though many more models are currently available.

Although promising, ECTIL requires further refinement. Because ECTIL only needs WSI-level labels, training datasets can be easily enriched with additional labelled data. For example, in our cohorts, only 170 (7%) of 2354 of samples were invasive lobular carcinomas, which probably contributed to the lower performance in this histological breast cancer. Additionally, using more recent foundation models can enhance representation of the underlying morphology. One of ECTIL's architectural limitations is that it treats patches independently and lacks context-awareness. Because context matters, future improvements could involve using context-aware models, such as vision transformers, which effectively handle WSIs. A vision transformer would capture spatial information and enables it to better learn to follow TILS Working Group scoring guidelines, such as excluding TILs present outside of the tumour bed and invasive margin.

Before a computational TIL assessment model such as ECTIL can be implemented in the clinic, a separate quality-control model is required, because even an empty glass slide image without tissue would otherwise be given a score, as occurs with existing models. Additionally, it is the responsibility of a pathologist and clinician to interpret the model's score and decide on the clinical outcome for the patient. Similar to current practice, in which a consensus of two pathologist scores is preferred, the consensus between AI and a pathologist should be considered best practice. Through finding consensus, the pathologist could identify, for example, severe pre-processing issues of the tissue or inclusion of incorrect areas by AI. Additionally, AI might identify regions that the pathologist overlooked due to their time constraints. Combined, we believe this would improve the scoring of tissue. Before implementing ECTIL as a direct pre-screening tool or a tool to assist the pathologist, prospective validation studies are required. Soon, ECTIL will be incorporated in a prospective de-escalation phase 3 clinical trial in patients with low-stage triple-negative breast cancer (NCT06476119).

In conclusion, ECTIL scores TILs on H&E FFPE WSIs in a single step without using deep learning-based segmentation and detection pipelines. ECTIL is data-efficient and compute-efficient and scores TILs highly similar to an expert pathologist. ECTIL also shows a similar HR in an overall survival analysis independent of clinicopathological variables in patients younger than 40 years with node-negative triple-negative breast cancer. In the future, such a computational TIL assessment model could be applied to pre-screen patients for prospective de-escalation or immunotherapy trials and to assist pathologists and clinicians in clinical decision making.

Contributors

YS, RV, MK, and HMH conceptualised the study. YS, RV, MO, RS, and HMH curated the data. YS and RM conducted formal analyses. YS developed methodology. YS, JT, and RM wrote software. YS and RV created visualisations. RV, MO, ML, GSS, GD, VdJ, YW, AGJvR, TGS, LCS, LGEdV, MvdV, PjvD, SL, JT, and RM provided resources. EJ, SL, JT, RM, MK, and HMH supervised the project. EG, JT, and HMH acquired funding. YS and

RV conducted the investigation. YS, RV, and MK drafted the original manuscript. YS and HMH validated the findings. HMH managed project administration. YS, RV, MO, RM, and HMH verified the raw data. All authors critically revised the manuscript and provided intellectual contributions. All authors reviewed and edited the manuscript. All authors had full access to all data, approved the final version for submission, and take responsibility for the content of the manuscript.

Declaration of interests

GSS reports research support to their institution from Agendia, AstraZeneca, Merck, Novartis, Pfizer, Roche, and Seagen. Consulting fees from Biovica, Novartis, and Seagen were paid to the providing institution, and GSS is a board member of the Dutch Medicines Evaluation Board (CBG). TGS received a payment or honoraria for podcasts outside the work submitted from Gilead Sciences. EGEDV reports institutional financial support for advisory boards or consultancy from NSABP, Daiichi Sankyo, and Crescendo Biologics, and institutional financial support for clinical trials or contracted research grants from Amgen, Genentech, Roche, Bayer, Servier, Regeneron, and Crescendo Biologics, all outside the submitted work. RS has undertaken a consulting or advisory role for Roche, AstraZeneca, BMS, and Daiichi Sankyo; is part of the speakers' bureau of Exact Sciences, AstraZeneca, and Daiichi Sankyo; received research support from Merck, Puma Biotechnology, and Roche; and received travel and accommodation expenses from Roche, Merck, AstraZeneca, and Daiichi Sankyo. PJvD has undertaken a consulting or advisory role for Visiopharm, Paige, and Sectra, and receives research funding from Paige, Pfizer, and Proscia. SCL has received grants or contracts from Roche, AstraZeneca, Eurocept Pharmaceuticals, Merck, Immunomedics (now Gilead), and Novartis; has undertaken a consulting or advisory role for Daiichi Sankyo, AstraZeneca, IBM, Cergentis (pro bono); and has received travel and accommodation expenses from Daiichi Sankyo Europe. MK has received research funding from BMS, Roche, and AstraZeneca; has undertaken advisory roles compensated to the institution for AstraZeneca, Daiichi Sankyo, Domain Therapeutics, Alderaan, BMS, MSD, and Roche; and has had a speakers fee compensated to the institution by Roche, BMS, and Gilead. All other authors declare no competing interests.

Data sharing

All the code required to train and validate the models is available online, including the scripts to reproduce the performance metrics and prognostic analyses. The whole slide images (WSIs) and clinical data of The Cancer Genome Atlas (TCGA) are publicly available online. The pathologist tumour-infiltrating lymphocyte (TIL) scores of TCGA samples are available online. Open access to the WSIs and clinical data of BASIS is available upon request online, and the pathologist TIL scores are available upon reasonable request. The datasets used and analysed in N4+ are not publicly available due to privacy-protection restrictions but are available from Prof S C Linn upon reasonable request and in the presence of a data transfer agreement (s.linn@nki.nl). The clinical data from MATADOR including TIL scores are publicly available online. The WSIs are not publicly available but can be requested for use from Prof S C Linn (s.linn@nki.nl) in the presence of a data transfer agreement. The data collected for TRAIN II can be made available to others in de-identified form after all primary and secondary outcomes have been published and in the presence of a data transfer agreement. Requests for data sharing can be made to Prof G Sonke, including a proposal that must be approved by the trial's steering committee (g.sonke@nki.nl). The datasets used and analysed during PARADIGM are not publicly available due to privacy-protection restrictions but are available from Prof S C Linn upon reasonable request and with permission from the Netherlands Cancer Registry, hosted by the Netherlands Comprehensive Cancer Center.

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For **Slide Score** see <https://www.slidescore.com/>

For the **TCGA Research Network** see <https://www.cancer.gov/tcga>

For the **training and validation code** see <https://github.com/nki-ai/ectil>

For **WSIs** see <https://portal.gdc.cancer.gov/>

For **clinical data** see <https://www.cbiportal.org/>

For **TIL scores** see <https://github.com/nki-ai/ectil>

For **WSIs and clinical data of BASIS** see <https://data.icgc-argo.org/>

For **MATADOR clinical data** see <https://data.mendeley.com/datasets/n24jpkp487/r2>

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